LazyLinear#

https://pytorch.org/docs/stable/generated/torch.nn.modules.linear.LazyLinear

Description:

A torch.nn.Linear module where in_features is inferred. In this module, the weight and bias are of torch.nn.UninitializedParameter class. They will be initialized after the first call to forward is done and the module will become a regular torch.nn.Linear module. The in_features argument of the Linear is inferred from the input.shape[-1]. Check the torch.nn.modules.lazy.LazyModuleMixin for further documentation on lazy modules and their limitations. out_features (int) – size of each output sample bias (UninitializedParameter) – If set to False, the layer will not learn an additive bias. Default: True

Function Signature

```
class torch.nn.modules.linear.LazyLinear(out_features,
bias=True, device=None, dtype=None)[source]#

Parameters

Variables

cls_to_become[source]#

initialize_parameters(input)[source]#
```

Detailed Documentation

Documentation

A torch.nn.Linear module where in_features is inferred.

In this module, the weight and bias are of torch.nn.UninitializedParameter class. They will be initialized after the first call to forward is done and the module will become a regular torch.nn.Linear module. The in_features argument of the Linear is inferred from the input.shape[-1].

Check the torch.nn.modules.lazy.LazyModuleMixin for further documentation on lazy modules and their limitations.

out_features (int) – size of each output sample

bias (UninitializedParameter) – If set to False, the layer will not learn an additive bias.

Default: True

weight (torch.nn.parameter.UninitializedParameter) – the learnable weights of the module of shape (out_features,in_features)(\text{out_features}, \text{in_features})(out_features,in_features). The values are initialized from U(−k,k)\mathcal{U}(-\sqrt{k}, \sqrt{k})U(−k,k), where k=1in_featuresk = \frac{1}{\text{in_features}}k=in_features1

bias (torch.nn.parameter.UninitializedParameter) – the learnable bias of the module of shape (out_features)(\text{out_features})(out_features). If bias is True, the values are initialized from U(−k,k)\mathcal{U}(-\sqrt{k}, \sqrt{k})U(−k,k) where k=1in_featuresk = \frac{1}{\text{in_features}}k=in_features1

Infers in_features based on input and initializes parameters.

Resets parameters based on their initialization used in __init__.

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